SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-016: Show/Hide Password Functionality in the Admin Add New Student Form

Field	Details
Change Request ID	CR-016
Change Request Title	Show/Hide Password Functionality in the Admin Add New Student Form
Student Name	T.Lakshika
Student ID	JF/ICT/24/32
Implementation Status	Completed
Date	9 October 2026

1. Change Request Summary

1.1 Business Requirement

The administrator needed a convenient way to view or hide the password entered in the Add New Student form. The change was required to reduce password-entry mistakes and improve the usability of the student registration interface.

1.2 Existing Behaviour

Before implementation:

- The administrator could enter a password in the Add New Student form.
- The password was displayed as hidden characters.
- There was no eye button to temporarily reveal the entered password.
- If the administrator made a typing mistake, it could be difficult to identify it before submitting the form.

1.3 Requested Change

A show/hide password feature was required in the Add New Student form.

The feature must:

- Display an eye icon inside the password input field.
- Hide the password by default.

- Show the entered password when the eye button is clicked.
- Hide the password again when the button is clicked a second time.
- Allow the administrator to toggle password visibility without clearing the entered password.
- Preserve the existing password validation and form submission functionality.
- Provide appropriate styling for the eye button.
- Avoid changing existing student creation and editing functionality.

2. Implementation

2.1 Implementation Summary

The password show/hide functionality was implemented in the Add New Student form using React state.

A Boolean state variable named `showPassword` controls the visibility of the password. When the administrator clicks the eye button, the state changes and the password input type switches between `password` and `text`.

The feature uses an inline SVG icon, so an additional icon library is not required for the eye button.

CSS styling was added to position the eye button inside the password input field and improve its appearance and usability.

2.2 Affected Files

File	Change	Purpose
`frontend/src/components/StudentFormModal.jsx`	Modified	Added password visibility state, eye button and toggle functionality
`frontend/src/App.css`	Modified / Verified	Added styling for the password input wrapper, eye button, focus state and validation messages
Additional files	None, if no other files were modified	No additional changes required

2.3 Technical Changes

Password Visibility State

A React state variable controls whether the password is visible.

```
const [showPassword, setShowPassword] = useState(false);
```

Initially, `showPassword` is `false`, so the password remains hidden.

Dynamic Input Type

The password input type changes according to the state.

```
type={showPassword ? "text" : "password"}
```

- `password`: hides the entered characters.
- `text`: displays the entered password.

Eye Button Functionality

The eye button changes the visibility state when clicked.

```
onClick={() => setShowPassword((previous) => !previous)}
```

The icon changes according to the current visibility state, helping the administrator identify the available action.

CSS Styling

CSS was added for:

- Positioning the eye button inside the password field.
- Preventing the text from overlapping the eye icon.
- Displaying a pointer cursor.
- Providing hover and keyboard-focus styles.
- Supporting the password field on smaller screens.

2.4 Implementation Logic

1. Open the Add New Student form.
2. Display the password input with hidden characters by default.
3. Display the eye button inside the password input field.
4. When the eye button is clicked, update the `showPassword` state.
5. Change the input type from `password` to `text`.
6. Display the entered password while the visibility option is enabled.
7. When the eye button is clicked again, change the input type back to `password`.
8. Preserve the entered value while switching between visible and hidden modes.
9. Continue the existing password validation and student creation process.

2.5 Error and Data Handling

The feature changes only the visibility of the password field. It does not change the entered password value.

The existing form validation and submission logic remain responsible for checking the password and sending the form data to the backend.

The password remains hidden by default whenever the form is initialized or reset, provided the component resets `showPassword` to `false`.

The eye button must not submit the form. It should use `type="button"` to avoid accidentally triggering form submission.

3. Testing & Verification

3.1 Testing Approach

The feature should be verified through manual frontend testing.

Testing Types:

- Functional testing
- Positive testing
- UI testing
- Password visibility testing
- Regression testing
- Form submission testing

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open the Add New Student form	Password field is displayed with hidden characters.	The Add New Student form opened successfully, and the password field displayed hidden characters by default.	Pass
TC-02	Enter a password	Entered characters remain hidden by default.	The entered password remained hidden in the password field.	Pass
TC-03	Click the eye button	Entered password becomes visible.	The eye button displayed the entered password when clicked.	Pass
TC-04	Click the eye button again	Password becomes hidden again.	The password was hidden again when the eye button was clicked a second time.	Pass

TC-05	Toggle visibility multiple times	Password value remains unchanged.	The password value remained unchanged when switching between visible and hidden modes.	Pass
TC-06	Submit a valid student registration form	Existing student creation process works correctly.	The student registration form submitted successfully, and the existing student creation process worked correctly.	Pass
TC-07	Submit an invalid password	Existing password validation still works.	The form displayed the appropriate validation error for an invalid password and prevented submission of invalid data.	Pass
TC-08	Click the eye button before submitting	Eye button does not submit the form.	Clicking the eye button only changed password visibility and did not submit the form.	Pass
TC-09	Check the layout on a smaller screen	Eye button remains properly positioned.	The eye button remained properly positioned inside the password field on a smaller screen without overlapping the input text.	Pass

Environment: Development

Browser: Google Chrome

Test Role: Administrator

3.2 Test Cases

3.3 Test Summary

Test Result	Count
Total Test Cases	9
Passed	9
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	Password is hidden by default	Open the Add New Student form	Met
AC-002	The eye button reveals the password	Click the eye button	Met

AC-003	The eye button hides the password again	Click the eye button a second time	Met
AC-004	Password value is preserved when toggling visibility	Toggle visibility several times	Met
AC-005	Existing password validation remains functional	Submit valid and invalid password values	Met
AC-006	Existing student creation functionality remains functional	Create a test student	Met
AC-007	Eye button does not submit the form	Click the eye button before submission	Met
AC-008	Eye button is appropriately styled and positioned	Inspect the form layout	Met

5. Evidence

A maximum of two relevant screenshots should be attached.

Evidence 01 – Password Hidden by Default

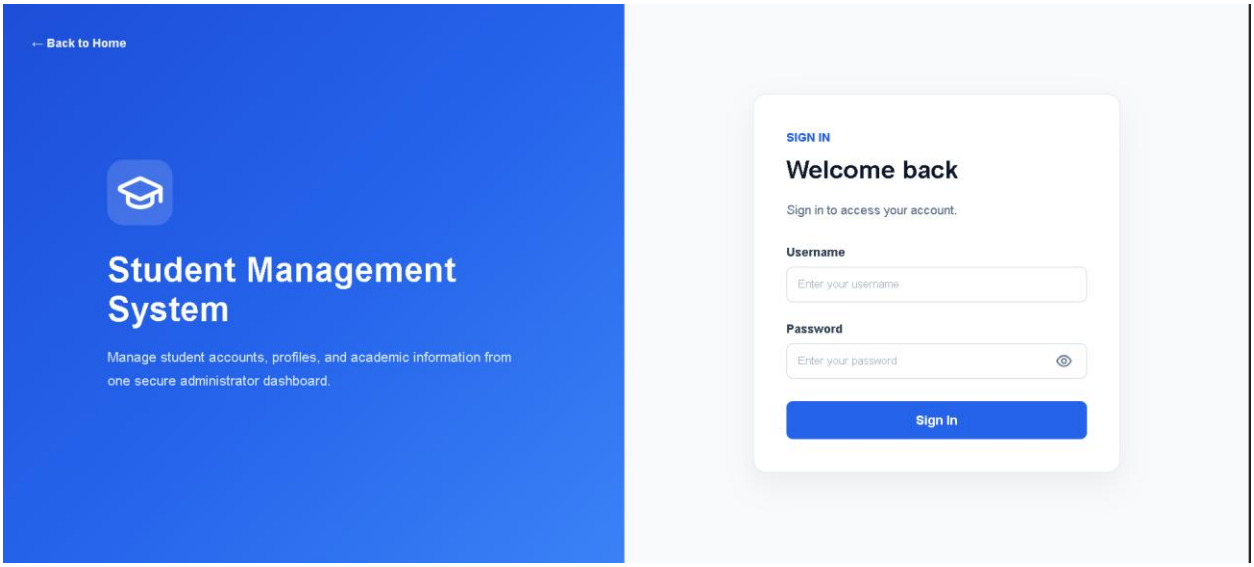
Related: AC-001, AC-008

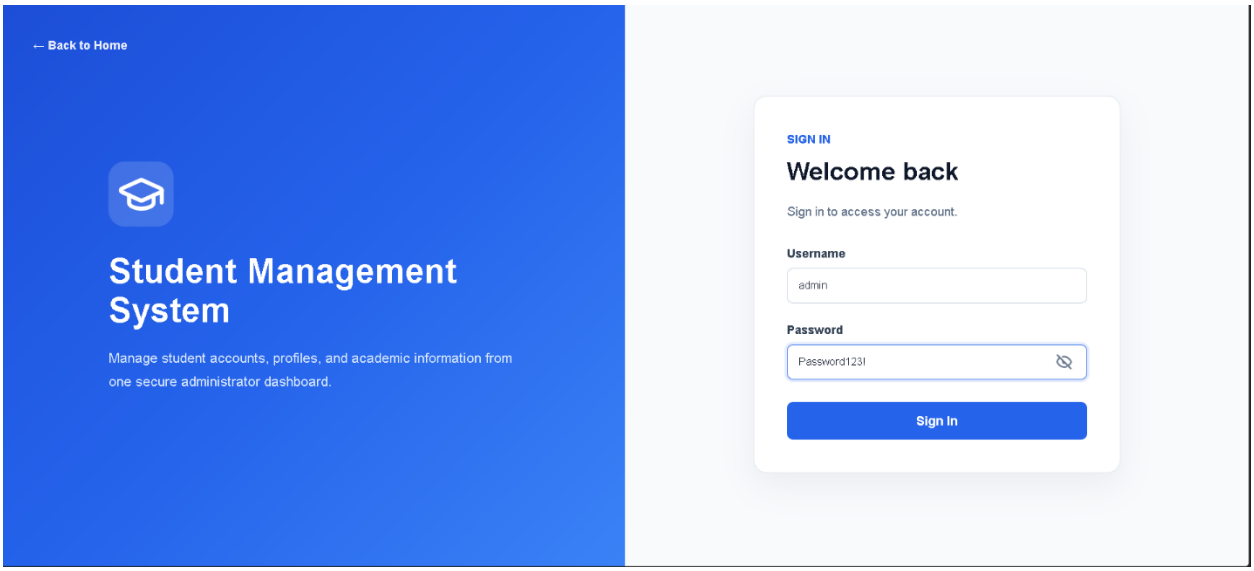
Figure 1: Administrator Add New Student form showing the password input with hidden characters and the eye button.

Evidence 02 – Password Visibility Enabled

Related: AC-002, AC-003, AC-004

Figure 2: Administrator Add New Student form showing the password visibility feature after clicking the eye button.





Add New Student

Create a new student account.

Registration Number
REG004

First Name
DAVI

Last Name
KUMARI

Email
davi@gmail.com

Phone Number
0711134564

Password
Davi123@

Password must contain at least 8 characters, including uppercase, lowercase, number and special character.

Cancel | Create Student

6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Existing Student Creation	Verified
Final Status	Accepted

6.1 Final Summary

The password show/hide functionality was introduced into the administrator's Add New Student form to improve password-entry usability. The feature uses React state to switch the password input between hidden and visible modes when the eye button is clicked.

The entered password value is preserved when visibility is toggled. CSS styling provides appropriate positioning and interaction feedback for the eye button. The implementation is intended to preserve the existing password validation and student creation process without requiring changes to the backend API or database.

The functionality should be accepted after the defined test cases and acceptance criteria have been verified.

6.2 Issues / Limitations

If testing identifies no issues:

No significant issues identified during testing.

If testing has not yet been completed:

Functional testing and regression verification are pending.

Student Name: T.Lakshika

Student ID: JF/ICT/24/32

Date: 9 October 2026

Signature: T.Lakshika

~ END OF REPORT ~